JULY-DECEMBER
2023

GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

"To prepare human resources with value-based competency for technical advancements and growth of society."

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**



ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standards of the society.**
- **To prepare engineers with effective communication skills, professional ethics and leadership qualities.**

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HOD'S MESSAGE



PROF. SANJAY D. JOSHI

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

Prof. Sanjay D. Joshi
HOD EC Department
Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

Program Outcomes (POs)

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

Program Outcomes (POs)

- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11: Life-Long Learning:** Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

DEPARTMENTAL DETAILS

ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Members



PROF. SANJAY D. JOSHI
DESIGNATION: ASSOCIATE PROFESSOR
EMBEDED SYSTEM, IMAGE PROCESSING, MICROCONTROLLER AND MICOPROCESSOR



DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP



PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS



DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS



PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION



PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY



PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN



PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS



PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR



L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.



ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench



L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.



L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)



ELECTRONICS AND COMMUNICATION ENGINEERING

Motivational Session by Start up Expert

Event : An Expert talk on"Motivational Session by Start up Expert on A Smart Home Automation Venture by Zavx"

11 Date: 21/10/2023

Time: 11:00 AM to 12:30 PM

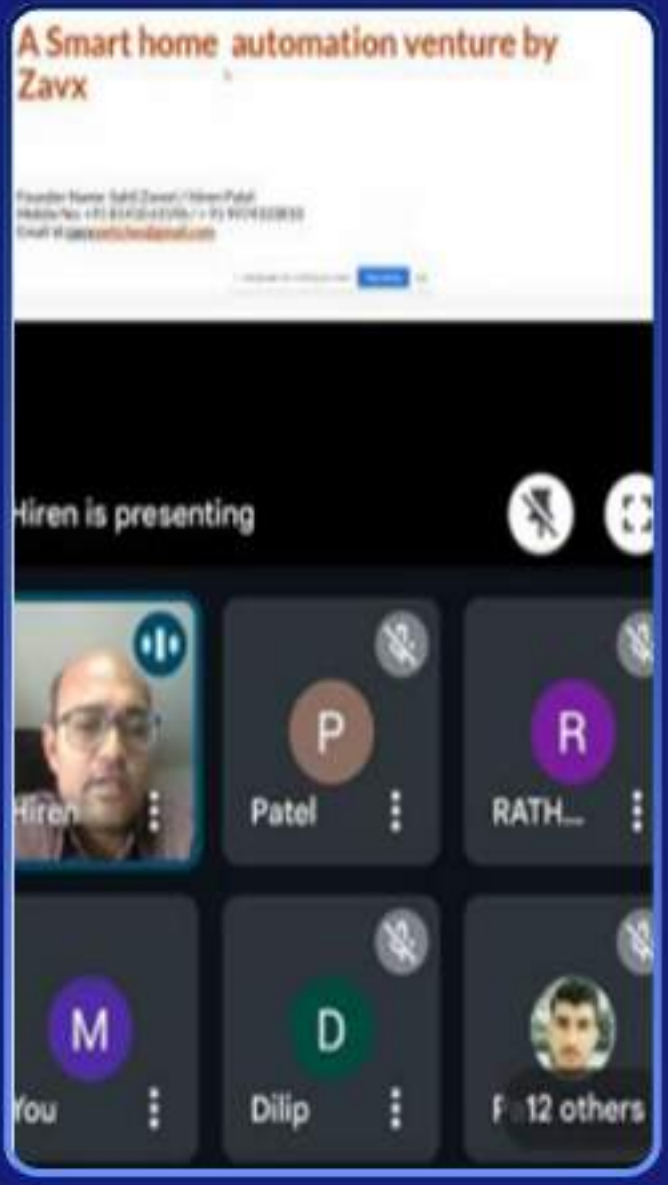
Platform: Google Meet

Objectives of Event:

Institution Innovation Council (IIC) an Initiative of MHRD for selected Higher Education Institutes. Ministry of Human Resource Development (MHRD), Govt. of India has established 'MHRD's Innovation Cell (MIC)' to systematically foster the culture of Innovation amongst all Higher Education Institutions (HEIs) Which helps in planning, channelizing and organising round the year activities in the areas of innovation, entrepreneurship and IPR to sensitise and motivate students and faculties to pursue innovation and start-up during their academic. Electronics and Communication Department has organised Impact Lecture on “Motivational Session by Start up Expert on A Smart home automation venture by Zavx " by an Expert Hiren Patel Ex.Assistant Professor of ECE Department , SVIT Vasad and Start up Enterprenure of Zavx. We have organised program at 21st October 2023 Saturday in the time duration of 11:00 A.M. to 12:30 P.M. by virtual mode of Google Meet. We have invited the student from all department and they have participated in the program. Speaker has given a so much valuable tips to student regarding Automation in Different Home Appliances , student start up program.Students have interacted with the speaker eagerly and encouraged to do the needful in the area of Automation and Start-up for having golden opportunity in Career growth

ELECTRONICS AND COMMUNICATION ENGINEERING

Glimpse of the Event









Guru Purnima Event organised on 3rd July 2023



On the 3rd of July 2023, the volunteers of NSS GEC Patan came together to havefun Guru Purnima, a widespread occasion that can pay tribute to our respected mentors.The festivities started out as all the volunteers amassed in front of the administrative block, filled with enthusiasm and appreciate. To honour the professors, the volunteers had been divided into groups to carry out the 'Tilak Ceremony' held in different departments. The volunteers took splendid pride in performing the ritual of doing 'Tilak' of every professor, symbolizing reverence and gratitude. Following this, in adherence to Indian way of life, the volunteers sought the blessings of every professor through the 'Charanasparsh' gesture, which entails touching their toes as a signal of respect and gratitude. The professors were deeply moved by the profound display of respect and appreciation, and they responded by imparting their heartfelt blessings and commending the volunteers for their dedication

'Guru Purnima' is a festival dedicated to offering respect to all the spiritual and academic guru. It is celebrated as a festival in India, Nepal and Bhutan by Hindus, Jains and Buddhists. This festival is traditionally observed to honour one's chosen spiritual teachers or leaders. It is observed on the full moon day (Purnima) in the month of Ashadha (June–July) according to the Hindu Calendar. The festival was revived by Mahatma Gandhi to pay tribute to his spiritual guru, Shrimad Rajchandra. It is also known as Vyasa Purnima, for it marks the birthday of Veda Vyasa, the sage who authoredthe Mahabharata and compiled the Vedas. As a part of these NSS GEC Patan organizedvarious programs so that we can give respect to our professors and to aware studentsabout the importance of Guru in our life



World Entrepreneur Day Celebration 22nd August 2023.

On 22 nd August 2023, Entrepreneurship event was organized by NSS Volunteers in order to promote economic growth, and provides access to goods and services. As a part of these NSS GEC PATAN had collaborated with Entrepreneur Club of GEC PATAN and had organized an event on World Entrepreneur Day to improves the overallstandard of living. The Dr. S & S. S. Gandhi Government Engineering College, Surat had organized an event for encouraging the Student Startup policy by giving them the token of encouragement, and the whole event was streamed live on the youtube channel of Dr. S.& S.S. Gandhi College YouTube which was made live for all the institutes of India . The event speaker was Dr. Sanjay R. Joshi and the co speaker was Mr. Gopal Arya. And the main speaker was respected Mr. Kashmiri Laal. The NSS Volunteers had made wonderful arrangement with the help of projector to watch the live stream together with all the students and faculty.

In this way, the students came to knew that their innovative ideas can get the nice place to be presented, and they get encouragement to do Startup and give their contribution to increase the economic of the India.

Major Outcomes of the Event :

- An Importance of startups among students had been spread.
- Contribution had been given in increasing number of employments.
- Helped in raising awareness about skill development and start-ups.



પાટણની સરકારી ઈજનેરી કોલેજ ખાતે સ્વતંત્રતા દિવસની ઉજવણી

(હમબૂઝ) પાટણ પાટણની સરકારી ઈજનેરી કોલેજના કેમ્પસમાં એન.એસ.એસ.ના વર્ષોપયોગી સપ્તાહ દરમ્યાન વિવિધ કાર્યક્રમ નું આયોજન

અભિયાન અંતર્ગત વકતૃત્વ સ્પર્ધા અને ચિત્ર સ્પર્ધા યોજી હતી જેના વિજેતાઓને પુરસ્કાર આપી પ્રોત્સાહિત કરવામા આવ્યા હતા.તથા કાર્યક્રમમાં એન.એસ.એસ.ના

કલ્પસર પ્રાથમિક શાળાના બાળ ઉત્સાહભેર જોડાયા હતા.

આમ, કોલેજના આચાર્ય અ એન.એસ.એસ.ના પ્રો ઓફિસરના સહકાર



Independence Day on 15th August, 2023 on the occasion to perform Flag Hoisting

Indian Independence Day is a momentous occasion that commemorates the freedom of India from British rule. It is celebrated on August 15th every year with great enthusiasm and patriotism. The day is marked by flag hoisting ceremonies, cultural events, parades, and patriotic songs. It is a time for Indians to reflect on the sacrifices made by freedom fighters and to celebrate the spirit of unity and independence. Our NSS Volunteer , faculties and students of primary school of Katpur celebrated 77th Independence Day with great fervor and patriotism at Government Engineering College Patan on August 15, 2023, We Sang our National Anthem and our Principal sir gives brief introduction of the importance of Independence day. We also conduct a competition on Drawing and Elocution , our Student gave inspiring speeches in Hindi , Gujarati , and English. A cultural program was also held in our college where college student and students of primary school are participated some student performed a group patriotic song, some student performs group dance and some student played a drama as a part of the occasion.

Major Outcomes of event

- **We learn the importance of our Indian culture by a Drama and Dance.**
- **We give awareness about our Freedom fighter to small child of katpur primary school.**
- **We build our unity and also gain much respect for our Nation.**





Chandrayaan 3 live streaming on 22 August 2023

On 23rd August 2023, Chandrayaan -3 Landing : NSS GEC PATAN had organised live streaming of ISRO's moon mission . It is a historic and proud moment for India. To be witness of this moment, NSS GEC PATAN had organised live streaming watch together event. Indian Space Research Organisation's (ISRO) Chandrayaan -3 moon mission was set to enter its final stage of moon landing on 23rd August, ISRO arranged livestream the launch that was began at 5:20 pm IST which was half an hour before the target landing time of 6:04 pm IST. This live telecast of the landing was streamed in GEC PATAN by NSS GECP Volunteers. At 5:20pm, this event was started . This event was attended by lots of colleagues and Professors. It was a really a very pride moment as a Indian to be a witness of such a milestone moment and felt that proud with their companions and colleagues . In this way , student and professors become the witness of such a historic milestone moment of successfully landing on the moon's far side after a month long journey through space. And felt proud with saluting to hardwork of our IS .O s

... ajo Outco..es of Event

• It was great experience for all to become .itness together of such a .isto ic mo..ent. • P oud ..omen. for all Indians and cheers fo our ISRO scientists.

• From these wa.ch together stude..s acknowledge about upcoming launches and projects of ISRO.







Cleanliness Drive (3.0) organized on 23 October 2023

DAY 1: Nr. Surroundings of Electrical Department

As a part of same Swachhta (3.0) drive was started from 23of October to 28 of October in which NSS Volunteers of our institutetook a initiative to clean college campus and surroundings in this drive,in which around 45 NSS Volunteers and Faculties joined together andcleaned the premises and compound of Electrical Department of college on the 1st day of the drive

DAY 2: Nr. Surroundings of Computer Department



As a part of Swachhta (3.0) drive, Our NSS volunteer has started this cleanliness drive from 23rd October to 28th October which took a initiative to clean college campus and surroundings. On the 2nd day of this drive, Around 45 NSS Volunteers and Faculties joined together and cleaned the premises and all the classrooms of Computer Department.All volunteers and faculties cleaned the area with full interest and hard-work and were dispersed with a great enthusiasm within them.



DAY 3: Nr. Surroundings of Civil

As a part of same Swachhta (3.0) drive was started from 23of October to 28 of October in which NSS Volunteers of our institutetooka initiative to clean college campus and surroundings in this drive,in which around 45 NSS Volunteers and Faculties joined together andcleaned the premises and compound of Civil Department ofcollege on the 3" day of the drive. All volunteers and faculties cleanedthe area with full interest and hard-work and were dispersed with angreat enthusiasm within them

Corruption Free India "organized on 1st November, 2023

As a Vigilance Awareness weak was going, under that Unit of NSS GEC Patan had organized a Corruption Free India event. On 1st November NSS Volunteers had organized this event in which professor, students and all the NSS Volunteers had actively took the participation. The speech competition on Corruption Free India was organized through which awareness about the how the corruption is spreading in all the fields now a days and also what is our role as an individual to stop the corruption and what step should we take to stop the corruption were discussed.

After the speech of all the students winner names declared thenat last pledge was given by all the students and professor thatwe will stop corruption and if we saw someone givingencouragement to the corruption we will definitely stop themand take the strict action against them



Swachta Hi Seva " organized on 1st November 2023

Electronics and Communication Department

Under Swachhtaa hi Seva, On 1st November, 2023, NSS Volunteers and faculties took a vow to clean our surroundings. We started with cleaning our campus and on 2nd day of drive we decided to clean E.C. department. We went to the E.C Department we cleaned the department and professors also joined us in this drive. They cleaned their cabins and faculty office, classrooms and labs helping us to clean department. Volunteer joyfully cleaned the surrounding with the help of workers/cleaners and Faculties. Some of the students from department also joined us to clean surrounding. They were happy and inspired by NSS Team works. After cleaning the area, volunteers, Students and professors washed their hand properly and sanitized. Our Principal and H.O.D appreciated our work. After this H.O.D sir called all the students and volunteer present there in the department and gave the speech on importance of clean surrounding and nation and also asked us to clean & maintain it daily.

Major Outcomes of The Event

- **We learned how to work in team.**
- **Importance of CLEAN SURROUNDING in our life.**
- **We learn to protect ourself from disease caused by surrounding garbage and trash.**
- **We learn importance of cleanliness for development of nation.**





World Remembrance Day for Road Traffic Victims" organized on 19th November 2023.







From Gujarat Road Safety Authority, on 19th November, 2023 an event was organized with the help of NSS Unit Cell GEC Patan to celebrate the World Remembrance Day for road traffic victims. In this event all the students and professor together had given their heartfelt tribute to that millions of people who have been and seriously injured on the world"s road and also acknowledging the suffering of all affected victims, families and communities

The host of the event had also shown the ratio of accident that happened per year. And gradually it is increased day by day, and it is now a time to take step to decrease the number of professor had taken the pledge that if they ever saw accident happened in front of them they will help the injured person and help them to went to the hospital or nearest clinic.

EDITORIAL TEAM

Patron

Dr. D. H. Patel

HOD, Electronics and Communication Department

Editors-in-Chief

Dr. M. B. Patel - Electronics & Communication Engineering

Content & Design Student Team

Joshi Parv Kashyap (240220111004 - EC)

Verma Samir Santoshbhai (240220111033 - EC)

Dahivat Ahir (250022011101 - EC)

Shiv Patel (250220111023 - EC)

Harsh Prajapati (250220111025 - EC)

Rushang Thakar (250220111034 - EC)